

Create PostScript And Netlist Using **XCircuit**

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With so many design tools and features, it is really difficult to identify which design tool to select without knowing about its features, working ability and advantages. Above all, the most important thing is to find out how easy the tool will be to use.

Many times, you come across a

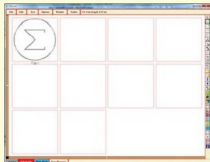


Fig. 1: Symbol example

situation where the tool is so complicated that learning it takes more than three months, while you have only four months to complete your design. Here is an attempt from EFY to help you understand XCircuit to an extent that you will be able to make the right choice.

Initially designed as a drawing program for undergraduate electrical engineers by Tim Edwards, XCircuit is now used worldwide not only for presentations but as an electronics design and automation (EDA) tool, too. Although the main purpose of XCircuit is to help create electric circuit models, it also allows you to create other two-dimensional schematics, like music typography, NAND gate, etc.

It makes the task of creating complex circuits easy with the use of some basic design elements, like circles,

lines and rectangles. These design elements can be rotated, flipped and joined to create desired circuit schematics that are used by the schematic capture to produce circuit netlists.

Unlike other design tools, XCircuit considers all circuits to be hierarchical, and thus provides hierarchical outputs. Edward says, "I use XCircuit probably more than any other tool, including for writing-out piano and violin scores and for writing Christmas cards."

Graphical programming advantage

Usually, what do you do? Create a circuit design using a design tool, write programming codes and then use the suitable format of codes in the circuit to check its working. In this process, you are required to learn a programming language (assembly-level or hardware-level) to start with, which is quiet time-consuming.

But, at the same time, if we have graphical design tools with which we do not have to spend time on coding, design graphics and the code for that will be auto-generated, like in case of MATLAB, it will be advantageous.

Designers get this opportunity with XCircuit that uses a PostScript file format for both input and output. PostScript is a programming language that gives publishable quality circuit diagrams as outputs. Unlike most graphical programs that provide output in a custom format (for example, .fig files for xfig) that can be translated to other graphical formats, PostScript is in ASCII format and need not be translated as it is widely-accepted.

This platform-independent programming format allows one to hand-

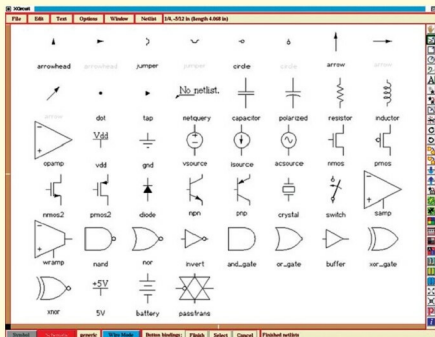


Fig. 2: Initial circuit window when starting XCircuit